

ControlPlane.ai

Finding it first — not finding out.

Business Proposal — Round 2, Accenture Innovation Challenge 2026

Team Muffin

1. Problem Framing

Consider a retail bank running a customer-service AI assistant. A customer asks about their loan's interest rate, and the assistant confidently quotes the wrong figure — or worse, includes another customer's account details in its answer, pulled from a poorly-scoped retrieval step. Neither failure is hypothetical: they are the two failure modes this concept was built around from Round 1, and they generalize directly to the brief's broader framing below.

Enterprises now run generative AI across many concurrent use cases at once — customer-facing chatbots like the one above, internal employee copilots, and decision-support tools embedded in regulated workflows. Each carries a different risk signature depending on the model behind it, the data it draws on, and how its output gets used downstream.

Today, that risk is discovered after the fact: after a customer acts on a wrong answer, after a regulator asks questions, after a leaked detail becomes a headline. The cost of that discovery gap compounds with scale — the brief's reference parameters assume tens of thousands of interactions per week across a mix of well-governed and loosely governed data sources, which means manual review cannot be the safety net.

The core problem ControlPlane.ai solves: enterprises need a way to catch confidently-wrong, expensive, or irresponsible AI behavior in real time, without slowing delivery down to a crawl or rebuilding a bespoke checker for every model and use case.

2. Solution Design

2.1 What it is

ControlPlane.ai is a model-agnostic monitoring layer that sits above any AI deployment and evaluates every response across three lenses — Performance (grounding / hallucination risk), Cost (compute and rework anomalies), and Responsibility (bias, safety, and privacy) — then takes a risk-tiered action rather than a single allow/block gate.

2.2 How it works

- Detection layer: fast, parallel, lightweight classifiers run on every response — grounding/retrieval verification, PII and entity detection, and cost/token anomaly detection — adding milliseconds rather than seconds.
- Escalation path: only borderline scores trigger a heavier secondary pass — a full 'AI-as-judge' LLM verifier or a deeper retrieval check — keeping the fast path cheap and the deep path reserved for genuine ambiguity.
- Decision logic: a weighted risk score maps to one of three tiers — Low (log silently, allow), Medium (auto-redact/edit and flag for review), or High (block and escalate to a human) — tuned per use case rather than one-size-fits-all.
- Governance layer: thresholds and weights are configurable per use case, geography, and risk appetite, so the same detectors can be tuned stricter for a regulated workflow and looser for low-stakes internal use, with a full audit trail behind every decision.
- Feedback loop: every human override (approve, escalate further, or dismiss) is captured and logged, forming the training signal that recalibrates thresholds and detector weights over time.

2.3 Why this shape

Because enterprises consume foundation models via API rather than owning them outright, the checker is designed to operate entirely at the input/output layer — it never needs access to model internals, which makes it portable across vendors and model versions. Because bias, hallucination, and privacy risks often overlap in a single response (a fabricated detail about a person is both a hallucination and a privacy concern), the design combines all three signals into one weighted score rather than forcing a brittle single-category classification.

3. Target Users

User	What ControlPlane.ai gives them
AI/ML engineering teams	Catch failures before users do; a single control layer instead of a bespoke checker rebuilt per model or use case.
Risk & compliance teams	A live, queryable audit trail instead of a postmortem; configurable policy per jurisdiction and regulation.
Business & product leaders	Confidence to greenlight AI rollouts in regulated, high-stakes settings — finance, healthcare, legal — where a silent failure is costly or dangerous.
Frontline reviewers	A tiered queue (not an all-or-nothing firehose) so attention goes to genuinely ambiguous cases, reducing alert fatigue.

4. Business Case & Impact

- Trust → adoption: turns 'hope it's right' into 'prove it's right, live' — commonly the single biggest blocker to enterprise AI rollout in regulated industries.
- Model-agnostic by design: sits above any model or vendor, so it is one control layer to maintain rather than a rebuild every time a team swaps models or a vendor ships a new version.
- Fast to pilot: lightweight classifiers mean a proof of concept can run alongside an existing AI deployment in weeks, not months, without touching production model infrastructure.
- Deliberately tuned, not falsely optimized: the alert-fatigue tradeoff (over-flagging vs. under-flagging) is treated as a design decision made explicit and adjustable per use case, rather than something the system claims to have 'solved.'

Illustrative cost framing: at a reference volume of tens of thousands of weekly interactions, even a small percentage of undetected high-risk responses represents meaningful liability exposure in regulated sectors — while a lightweight, parallelized detection layer adds negligible marginal compute cost per interaction compared to the cost of a single serious incident (regulatory, reputational, or customer-trust).

5. Phased Roadmap

Phase	Focus	Key deliverables
Phase 1 (0–2 mo)	Pilot on one use case	Deploy fast classifiers (grounding heuristic, PII/entity detection, cost baseline) on one contained use case; establish audit trail and override capture; calibrate thresholds against real traffic.
Phase 2 (2–5 mo)	Multi-use-case rollout	Extend to all three use-case archetypes with per-use-case policy configuration; introduce the heavier AI-as-judge

		verifier pass for borderline cases; build the reviewer queue UI.
Phase 3 (5–9 mo)	Governance & scale	Add jurisdiction-specific policy packs; build the recalibration feedback loop from human overrides into detector thresholds; add monitoring dashboards for false-positive/negative rates reported to stakeholders.
Phase 4 (9+ mo)	Platform expansion	Support multi-turn and agentic workflows where one questionable output can shape several downstream actions; expand retrieval-verification sources; explore cross-enterprise benchmark sharing for detector quality.

6. Key Risks & Mitigations

Risk	Mitigation
Over-flagging causes alert fatigue and reviewers start ignoring warnings.	Tiered actions (not a single gate) plus per-use-case tunable thresholds; the alert-fatigue tradeoff is monitored explicitly via false-positive/negative tracking, not assumed away.
Under-flagging misses a real incident (higher liability than a false alarm).	Default bias toward escalation under uncertainty; borderline scores always route to a heavier verifier pass rather than being auto-allowed.
No reliable real-time ground truth to check some claims against.	Grounding checker degrades gracefully — uses source-overlap when a reference document exists, and an explicit, labeled assertiveness heuristic when it doesn't, rather than silently guessing.
Rigid, hard-coded rules age quickly as regulation evolves.	Policy layer is configuration, not code — thresholds, weights, and jurisdiction-specific rules are adjustable without redeploying the detection logic.
Multi-turn / agentic use cases compound risk across turns.	Roadmap Phase 4 explicitly scopes conversation-level and action-level risk tracking rather than treating each response as independent.
Enterprises only expose model input/output, not internals, limiting inspection depth.	By design, all detection operates at the input/output layer, so this is a constraint the architecture is built around rather than something it depends on avoiding.

7. Prototype Scope (This Submission)

The accompanying prototype demonstrates the core detection-and-decision mechanism end-to-end on simulated data: three parallel detectors (grounding, cost anomaly, PII/policy), a weighted tiered decision engine, per-use-case configurable governance thresholds, an audit trail with human-override capture, and a metrics dashboard including a live false-positive/false-negative tradeoff simulator. See the repository README for the full mapping between each brief requirement and how the prototype addresses it, and for what would change between this proof-of-concept and a production build.

8. Submission

Submitted by Team Muffin for Round 2 of the Accenture Innovation Challenge 2026, Problem Track 1:
ControlPlane.ai.